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Palaeomagnetic and chronological study of Miocene welded tuffs in the northern part of Central Japan: tectonic implications for the latest stage of arc formation of Japan

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Abstract

Kinematical adjustment of tectonic blocks has occurred in the latest stage of arc formation of Japan within central Japan. Miocene welded tuffs have been sampled for chronological and palaeomagnetic studies from the northern part of central Japan. Zircon fission-track dating confirms Middle to Late Miocene ages (13.7–5.0 Ma) for the Yatakezawa, Shiobara, Kawaji, Kinugawa, Katashinagawa, Okkai and Ashio Formations. Characteristic directions with high unblocking temperature component above 560°C are isolated from 36 sites. Palaeomagnetic directions after tilt correction provide a Miocene characteristic palaeomagnetic direction of the northern part of central Japan (36.8°N, 139.3°E): $D = 7.3°$, $I = 50.3°$, $\alpha_{95} = 6.6°$. Presence of normal and reversed polarities, and a fairly long time average exclude the effect of geomagnetic field secular variation. Comparison with the Miocene palaeomagnetic poles of the neighbouring blocks indicates that the northern part of central Japan and northeast Japan behaved as a unified block since 15 Ma, whereas the Kanto Mountain region moved kinematically as an independent block from the northern block comprised of central Japan and northeast Japan at least between 16 Ma and 12 Ma. The boundary between the northern part of central Japan and the Kanto Mountain was in a convergent regime with dextral strike slip. A southward force exerted from the northern part of central Japan, as well as a northward force by the Izu-Bonin ridge, induced the clockwise rotation of the Kanto Mountain region. The northern part of central Japan had welded completely with the Kanto Mountain region by 6 Ma.

Keywords: tectonics; palaeomagnetism; chronology; Japan arc; Miocene

1. Introduction

Building of the present arc configuration of the Japanese islands has been recognized as a consequence of differential tectonic rotation of southwest and northeast Japan (e.g., Otofuji et al., 1985a; Faure and Lalevee, 1987; Jolivet et al., 1994). Associated with the opening of the Japan Sea, a unit block of northeast Japan was rifted from the Asian continent and rotated counter-clockwise through about 46° since 20 Ma, whereas southwest Japan rotated clockwise through more than 45° between 20 Ma and 14 Ma (Otofuji and Matsuda, 1987; Otofuji et al., 1991, 1994; Jolivet et al., 1994). Consequently, the southwestern part of northeast Japan and the

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northeastern part of southwest Japan had met along the central part of Japan and the formation of the proto-Japan Arc continued until 14 Ma.

The occurrence of a kinematical adjustment of tectonic blocks in central Japan has been speculated during the last decade (Hirooka et al., 1990; Kanaori, 1990; Hayashida, 1994; Fournier et al., 1995). Palaeomagnetic studies have identified a large clockwise rotation during a period later than 16 Ma for the Kanto Mountain region (Hyodo and Niitsuma, 1986). The continuation of this clockwise rotation later than 12 Ma was suggested with the palaeomagnetic evidence of Miocene volcanic products (Takahashi and Watanabe, 1993). Tectonic studies on relative rotation of blocks within central Japan have provided a clue for depicting an important aspect of the detailed arc formation in the latest stage of the present Japan Arc. The block rotation emphasizes the need to establish the Miocene palaeomagnetic directions for blocks within central Japan.

The Uetsu–Ashio zone occupies a fairly wide area in central Japan (Fig. 1), which is located to the northeast of the Kanto Mountain region. We choose this zone as a tectonic block in central Japan to study relative block rotation on the basis of palaeomagnetism. This study focuses on the Miocene welded tuff formations. We present new chronological results by the fission-track method as well as a palaeomagnetic study of the Middle to Late Miocene welded tuff sequences, because few chronological results have been reported from central Japan.

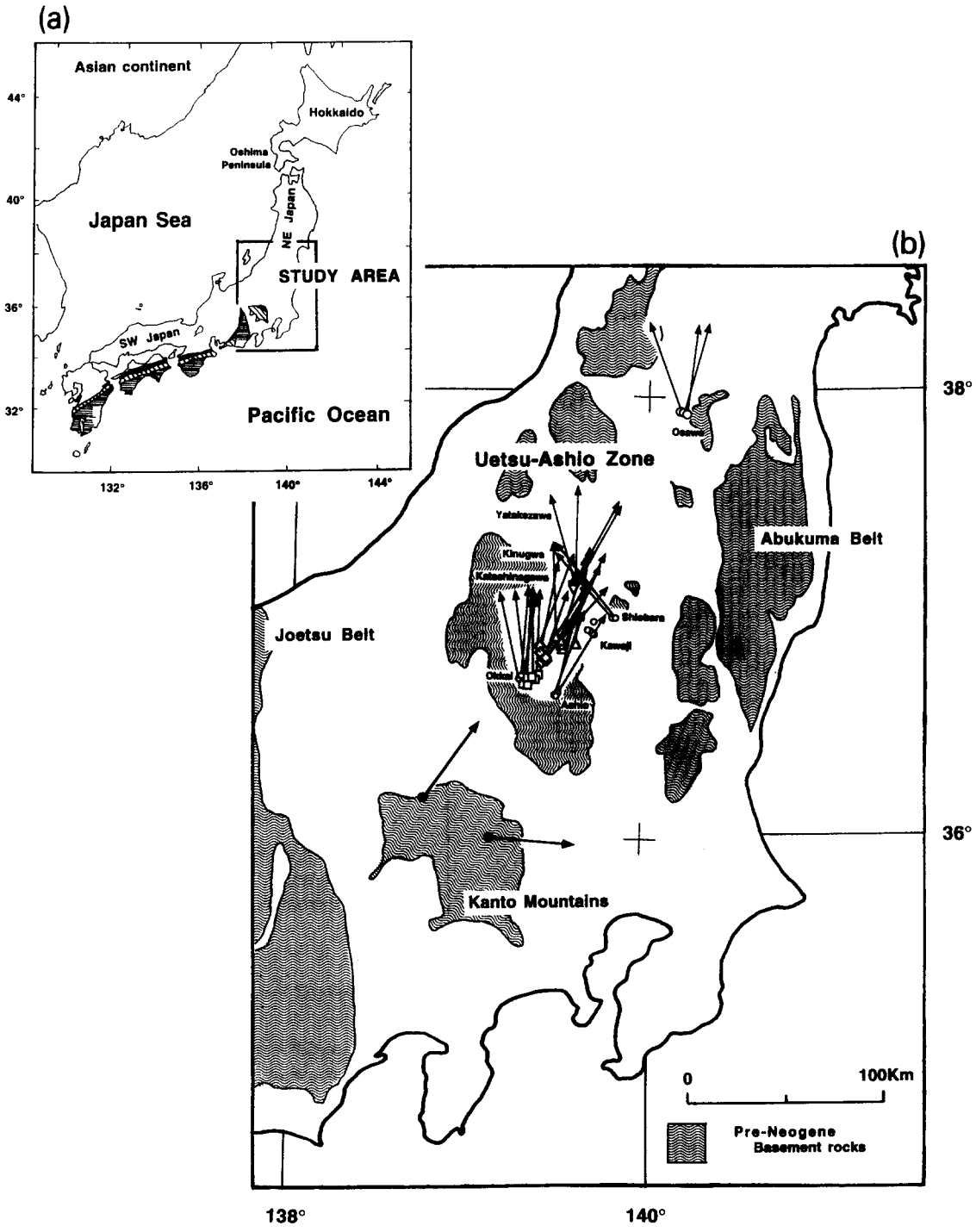
2. Geological setting and sampling

Central Japan is the most complicated geological segment in the Japanese islands because it is now located at the junction of four plates (Pacific, Philippine, North America, and Eurasia plates) (Demets, 1992) and was a boundary area in which the northeast and southwest Japan arcs had been welded in Miocene times (e.g., Taira et al., 1989; Otsuki, 1990) (Fig. 1).

The basement of the northern part of central Japan consists of Mesozoic to Palaeozoic rocks comprising the Joetsu and Abukuma metamorphic belts, and the Uetsu–Ashio zone (Editorial Committee of Kanto, 1986; Hashimoto, 1991). Since the latter zone has preserved Palaeozoic–Mesozoic sequences (e.g., the occurrence of Fusulinid-bearing limestone) quite similar to the Mino–Tamba terrane in southwest Japan (Isozaki et al., 1990), this zone is also recognized to be the Jurassic accretionary prism. The southern part of central Japan is occupied by Jurassic to Miocene basements composed of the Inner zone (Ryoke Belt) and the Outer zone (Sambagawa Belt, Chichibu–Sanbozan Belt, Shimanto Belt) (Geological Survey of Japan, 1982). The zonally arranged terranes of the two zones are an eastward extension of the geological structure of southwest Japan, although a significant change is observed in the trend of the lineation of the zonal structure between central and southwest Japan.

Intense felsic igneous activity occurred during Cretaceous to Eocene time in the Mesozoic to Palaeozoic terranes of central Japan (Fig. 2) (Editorial Committee of Kanto, 1986). The Okunikko Rhyolites whose ages have been determined at 56~65 Ma by K–Ar dating (Yanai, 1972) were formed during this Palaeogene activity in the Ashio zone in the northern part of central Japan. Bimodal volcanism followed in Middle Miocene times (Sato, 1994), so that thick piles of volcaniclastics were produced from the Ashio to Yonezawa region (Table 1): the Osawa Formation in the Yonezawa area, and the Yatakezawa, Shiobara, Kawaji, Kinugawa, Katashinagawa, Okkai and Ashio Formations in the Ashio area. Volcanism during Late Miocene

Fig. 1. (A) Sketch map of the study area, and configuration of the Japan arc and the zonally arranged Outer zone of southwest Japan. A significant change is observed in the trend of the lineation of the zonal structure of the Outer zone between central Japan and southwest Japan. (B) Map of the study area of central Japan showing location of sampling sites for the palaeomagnetism and chronology. Arrows indicate the palaeomagnetic directions for the sampling sites (squares: southwestern part of the Katashinagawa Formation; diamonds: northeastern part of the Katashinagawa Formation; triangle: the Kinugawa Formation; inverse triangle: the Yatakezawa Formation; circles: the Osawa, Shiobara, Kawaji, Ashio and Okkai formations). The bold arrows mark the palaeomagnetic directions previously found from 16 Ma sediments and 12 Ma volcanic products in the Kanto Mountain region (Hyodo and Niitsuma, 1986; Takahashi and Watanabe, 1993).


(a)
Asian continent
Hokkaido
Oshima Peninsula
NE Japan
Japan Sea
STUDY AREA
SW Japan
Pacific Ocean
44°
40°
36°
32°
132°
136°
140°
144°
(b)
Uetsu-Ashio Zone
Abukuma Belt
Joetsu Belt
Kanto Mountains
38°
36°
138°
140°
0
100Km
Pre-Neogene Basement rocks

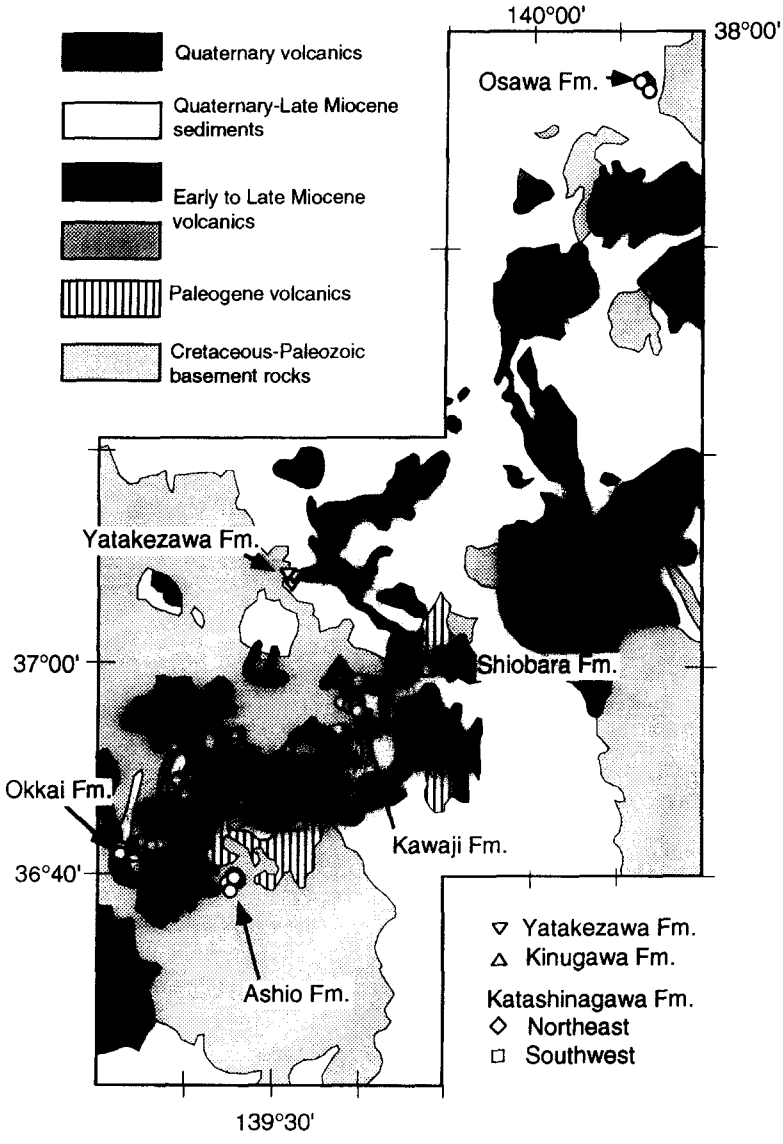


Fig. 2. Schematic geological map of the study area after Geological Survey of Japan (1982), showing location of sampling sites.

to Pliocene times was less intense, but intensive volcanism was re-initiated from the latest Pliocene to Quaternary times. Consequently extensive areas of the Palaeogene to Miocene formations are covered with Quaternary pyroclastics, and the Middle Miocene formations are therefore now exposed sporadically in central Japan.

We focus the palaeomagnetic study on the welded tuffs in the Middle to Late Miocene volcaniclastics distributed from 36°39′N to 37°56′N in latitude and from 139°13′E to 140°14′E in longitude (Figs. 1 and 2). More than 400 samples were collected for palaeomagnetism from 36 sites of the above-mentioned formations in the Yonezawa and Ashio areas. These areas have been geologically mapped in detail (Kusanagi, 1957; Kawada, 1966; Yanai, 1972; Sudo, 1976; Kitamura, 1986; Editorial Committee of Kanto, 1986; NEDO, 1986; Editorial Committee of Tohoku, 1989). Although a small number of radiometric ages were reported for these for-

Table 1
Summary of a stratigraphic column for the Ashio area

Area / Period/Epoch	Katashimagawa Area & Ashio Mountain	Kawaji & Shiobara Areas
Pliocene	Kinugawa Formation	Kinugawa Formation
Miocene	Okkai Formation Ashio Formation	Yatakezawa Formation Kawaji Formation Shiobara Formation
Paleogene & Cretaceous	Katashinagawa Formation Okunikko Rhyolites	
	Sudo (1976) *Yanai (1972)*	*NEDO (1986)* *Kitamura (1986)*

The stratigraphic column is based on Sudo (1976) for the Katashinagawa area, Kusanagi (1957), Yanai (1972) for the Ashio Mountain, and NEDO (1986) and Kitamura (1986) for the Kawaji and Shiobara areas.
Kinugawa Formation: 1.4 to 6.0 Ma (NEDO, 1986): K-Ar (whole rock)
Yatakezawa Formation: 10.0, 10.1 Ma (NEDO, 1986): K-Ar (whole rock)
Kawaji Formation: 10 to 15 Ma (Kitamura, 1986): FT (zircon)
Okunikko Rhyolites: 56 to 65 Ma (Yanai, 1972): K-Ar (biotite)

mations (Table 1) (Yanai, 1972; Kitamura, 1986; NEDO, 1986), we hardly ascertained reliability in their results because their ages were presented without detailed analyzing data. Samples for fission-track dating were therefore collected from seven formations except the Osawa Formation.

The welded tuffs were collected as hand samples oriented by magnetic compass for palaeomagnetism. Each palaeomagnetic site typically comprised twelve independently oriented samples, distributed over distances ranging up to 15 m. The eutaxitic structure (lineation of stretched pumice and aligned phenocrysts) was identified at each outcrop of the welded tuffs and this was used to make a correction for post-emplacement tilting.

3. Age determination

Fission-track (FT) dating was conducted for zircons from sixteen welded tuff samples in the seven formations. Zircon ages were determined by the external detector method using uranium-free muscovite (e.g., Naeser, 1976). Zircons of 60–100 mesh size were separated using conventional magnetic and heavy-liquid techniques. Zircon separates were mounted in PFA Teflon sheets, ground and polished to reveal 4 π (internal) surfaces. Etching of zircons took place in molten KOH and NaOH eutectic at either 225°C for times ranging from 15 to 32 h or 245°C from 6 to 14 h. The etched zircons were irradiated with standard glass covered by the muscovite external detectors in either the graphite or D_2O facility of the Kyoto University Reactor, for which Cd-ratios for Au are >2000 and 400, respectively. After irradiation, the external detectors were etched by 46% HF for 20 min at 20°C. The FT densities were measured at a magnification of 925× under a dry 100× objective lens. Ages were calculated using the zeta calibration approach. We used a zeta-value of 386.3 ± 23.0 determined dosimeter glass SRM 612 using the Fish Canyon Tuff, Buluk member tuff II and Tardree Rhyolite (Hurford, 1990). Errors were calculated using the 'conventional analysis' given by Green (1981). Uncertainties in the chronological data are quoted at 2 sigma. Results of the FT dating are listed in Table 2.

Only five of the sixteen samples passed the $P(\chi^2)$ test. Since all block samples for the FT dating contain small amounts of lithic fragments, the analyzed zircon grains are expected to be partly contaminated by detrital zircon grains. Low $P(\chi^2)$ are presumably ascribable to contamination of the detrital grains. The central FT age and age dispersion (Galbraith and Lasett, 1993) are calculated for individual samples to evaluate the degree of the contamination (Table 2). The five samples, which passed the $P(\chi^2)$ test, reveal

Table 2
Results of zircon fission-track analysis using the external method

Site	Sample	n	N_s	ρ_s	Ni	ρ_i	Nd	ρ_d	P(χ^2)%	Pooled age (Ma) $T \pm 2\sigma$	Central age (Ma) $T \pm 2\sigma$	Mean track length $L \pm \sigma$ (μm)	n^*
Yatakezawa Formation													
KN55	KN-55a	22	3119	2.29	10583	7.78	6105	1.75	<0.5	9.9 ± 0.5	**9.5 ± 0.7**	11.6 ± 1.3	193
Shiobara Formation													
KN51	KN-51a	25	2702	1.45	17333	9.36	6105	1.82	<0.5	5.5 ± 0.3	**5.0 ± 0.5**	11.4 ± 0.9	57
Kawaji Formation													
KK65	KK-65a	16	1721	0.98	4359	2.48	6078	0.956	<0.5	7.3 ± 0.5	**6.8 ± 0.7**	11.0 ± 1.2	36
Kinugawa Formation													
KN61	KN-61a	19	1495	0.801	6662	3.57	6105	1.68	0.7	7.3 ± 0.5	**6.7 ± 0.7**	11.7 ± 0.8	46
KN81	KN-81a	36	1215	0.591	6604	3.21	5926	2.20	10	**7.8 ± 0.5**	7.1 ± 0.6	11.9 ± 0.7	39
Katashinagawa Formation													
Northeast													
KK24	KK-24a	20	3900	1.77	7776	3.54	6078	0.955	0.5	9.2 ± 0.4	**8.9 ± 0.5**	11.5 ± 1.0	46
KK12	KK-12a	18	3597	1.88	6877	3.59	6816	0.933	<0.5	9.5 ± 0.5	**9.1 ± 0.6**	11.1 ± 1.4	59
KK20	KK-20a	20	3816	1.71	9166	4.10	7246	0.990	<0.5	8.8 ± 0.6	**7.7 ± 0.5**	10.3 ± 1.5	48
KN21	KN-21a	25	5238	2.50	11945	5.69	6658	1.10	<0.5	9.3 ± 0.4	**8.9 ± 0.6**	11.6 ± 0.7	135
Southwest													
KK02	KK-02a	7	2484	6.66	1350	3.62	6078	0.957	40	**33.8 ± 2.4**	33.5 ± 2.2	9.3 ± 2.2	68
KK08	KK-08a	14	3079	3.96	3303	4.25	6816	0.892	<0.5	16.0 ± 0.9	**13.3 ± 2.6**	8.7 ± 1.6	35
KK14	KK-14a	13	1951	2.76	6412	9.06	5926	2.33	15	**13.7 ± 0.8**	13.5 ± 1.0	11.9 ± 0.9	43
KK16	KK-16a	15	1540	1.34	2572	2.23	6308	1.04	10	**12.0 ± 0.8**	11.6 ± 0.8	11.5 ± 0.6	30
Okkai Formation													
KK04	KK-04a	18	1211	1.61	2127	2.83	6078	0.956	1	10.6 ± 0.9	**9.8 ± 1.1**	11.3 ± 1.0	33
KK18	KK-18a	24	3152	0.992	4809	1.51	6308	0.866	5	**10.9 ± 0.6**	10.6 ± 0.6	11.1 ± 0.9	35
Ashio Formation													
KN15	KN-15a	28	6901	2.32	11691	3.93	6122	0.957	<0.5	10.9 ± 0.4	**10.2 ± 0.8**	11.5 ± 0.9	117

Fission-track age and mean track length are listed. Boldface in the fission-track ages are reliable ages or first-order approximation of ages for the samples (see text). n = number of counted grains. N = number of tracks counted; subscripts: s for spontaneous; i for induced; and d for dosimeter. Track densities (ρ) are $\times 10^6$ tracks cm^{-2}; subscripts: s for spontaneous; i for induced; and d for dosimeter. P(χ^2)% is probability obtained by the χ^2 method for N degrees of freedom (where N is $n - 1$) (Galbraith, 1981). Pooled age = fission-track age calculated from pooled ρ_s and ρ_i for all grains counted with a ±2 sigma Poisson error. Central age = fission-track ages calculated from weighted mean ages of grains with a ±2 sigma error (Green, 1981; Galbraith and Lasett, 1993). L = mean track length. n^* = number of tracks counted. Locations of sampling sites are listed in Table 3.

that the central FT ages and their ±2 sigma age dispersion are quite similar to values for the pooled ages (e.g., KK-14a; 13.5±1.0 vs. 13.7±0.8). In contrast, the remaining samples, which failed the test, show that the central FT ages are slightly younger than the pooled ages and that their dispersions are larger than the errors of the pooled ages (e.g., KK-08a; 13.3 ± 2.6 vs. 16.0 ± 0.9). This confirms our expectation that the contamination of detrital zircon grains occurred in the samples with low P(χ^2) value. However, the dispersion of the central age for each sample is small, suggesting that the amount of detrital zircon is not large. This implies that the measured FT ages are close to the true depositional ages. We use the pooled FT ages as reliable ages for samples which passed the P(χ^2) test, whereas we recognize the central FT ages as a first-order approximation for ages of the samples which failed in the test.

Spontaneous track length distributions were analyzed to evaluate the possible effect of the thermal events on the FT ages, because volcanic activity has continued until the Quaternary in the studied area (Editorial Committee of Kanto, 1986). Although our samples appeared to have suffered little alteration,

the fission track lengths may have been shortened during later volcanic events. The lengths of horizontally confined FT were measured under a 200× dry objective lens, using a video-camera-display system and a digitizing tablet interfaced with a microcomputer. Except for two samples (KK-02A, KK-08a) from the southern part of the Katashinagawa Formation, almost all samples revealed long mean track lengths (<11 μm) with small standard deviations. Therefore it is unlikely that these zircon FTs have been shortened since their deposition. Mean track lengths around 9 μm are observed in the samples of KK-02a and KK-08a, implying that they were brought into the zircon partial annealing zone (probably $\sim 240 \pm 50$°C, Hurford, 1986) some time after their deposition, so that these data are excluded from further tectonic investigation.

The FT ages assign a period between 13.7 Ma and 5.0 Ma to deposition events for the Yatakezawa, Shiobara, Kawaji, Kinugawa, Katashinagawa, Okkai and Ashio Formations. Reliability of the FT age for the Yatakezawa Formation may be ascertained through agreement with the K–Ar ages within the error of 2 sigma (10.0, 10.1 Ma by the K–Ar age vs. 9.5 ± 0.7 by the FT age) (NEDO, 1986). The present study provides different ages from the previously reported age data for the Kinugawa and Kawaji Formations (Table 1) (Kitamura, 1986; NEDO, 1986). Because no detailed information on analyzing results was reported in previous works, farther discussion is outside our scope. Our FT ages are consistent with stratigraphic estimates (Editorial Committee of Kanto, 1986). We conclude that felsic volcanism was active during Middle to Late Miocene times in the northern part of central Japan.

The FT ages of the Katashinagawa Formation are divided into two groups: the younger ages are observed from the four samples in the northern part of the formation (7.7–9.1 Ma), whereas the southern part reveals the older ages of 13.7 and 12.0 Ma. This indicates that the Katashinagawa formation formed during two different volcanic activities: the older activity at 12–13 Ma and the younger one at 8–9 Ma. A rock sample of site KK02 showed the oldest age of 33.8 Ma in this formation, suggesting that this may belong to the volcanic unit of the preceding volcanic activity of, e.g., the Okunikko Rhyolites which are sporadically distributed in the Katashinagawa Formation (Editorial Committee of Kanto, 1986).

4. Palaeomagnetism

In the laboratory, individual specimens, 25 mm in diameter and 23 mm long, were prepared from these samples. Natural remanent magnetizations (NRMs) were measured with either a spinner magnetometer (Natsuhara SMM-85) or a ScT cryogenic magnetometer according to the intensity of the magnetization. All specimens were subjected to stepwise thermal demagnetization up to 620°C. Results for each specimen were plotted on orthogonal vector diagrams (Zijderveld, 1967) to assess component structure as well as on equal-area projections to evaluate directional stability. Principal component analysis (Kirschvink, 1980) was used to estimate the component directions. Palaeomagnetic results are listed in Table 3.

4.1. Demagnetization

Progressive thermal demagnetization of the samples separates the 36 sampling sites into three categories (Fig. 3).

Category A. Sixteen sites exhibit a single magnetic component with unblocking temperatures of 600–630°C (Fig. 3: 1–3). Little change is observed in the direction of remanent magnetization after demagnetization at temperatures higher than the Curie temperature of magnetite. Both magnetite and hematite probably carry the stable remanent magnetization with almost the same direction.

Category B. Fifteen sites reveal two distinct magnetization components (Fig. 3: 4–6). The unblocking temperature of the high-temperature component ranges between 590°C and 630°C, indicating that both magnetite and hematite carry the stable remanent magnetization. The low-temperature components are removed in the initial stage of demagnetization by 300°C.

Category C. Three components have appeared in four sites of the Kawaji Formation and in one site of the Ashio Formation (Fig. 3: 7 and 8). The lowest-temperature component is removed by 250°C from all sites for both formations. After the middle temperature component is erased at 350°C for the Kawaji

Table 3
Summary of palaeomagnetic results for the Miocene welded tuffs from the northern part of central Japan: high-temperature component

Site	Category	N	In situ		Tilt corrected				Locality		Rock type	Age (Ma)
			D(°)	I(°)	D(°)	I(°)	k	α_{95} (°)	Lat. (°N)	Long. (°E)		
Osawa Formation												15.5[a]
RY29	A	11	0.3	46.8	338.1	35.6	275.1	2.8	37°55.7′	140°12.4′	RWT	
RY31	A	12	348.1	47.7	14.3	48.4	449.6	2.0	37°55.2′	140°13.5′	RWT	
RY67	A	10	349.2	35.0	6.2	37.2	55.5	6.5	37°55.7′	140°12.4′	RWT	
Yatakezawa Formation												
North												
KN55	A	12	5.8	67.0	343.9	59.7	363.5	2.3	37°08.3′	139°32.2′	DWT	9.5
KN57	A	11	4.5	65.7	0.4	78.6	851.9	1.6	37°08.5′	139°31.9′	DWT	
South												
KN59	A	12	345.5	65.0	27.8	56.7	277.3	2.6	37°07.9′	139°31.9′	DWT	
KN96	A	9	20.5	60.4	28.0	38.2	244.8	3.3	37°07.9′	139°31.9′	DWT	
Shiobara Formation												
KN51	B	12	134.4	−31.4	139.7	−46.0	133.8	3.8	36°59.9′	139°47.3′	DRWT	5.0
KN53	B	12	117.0	−33.0	138.1	−43.2	625.1	1.7	36°59.8′	139°47.4′	DRWT	
Kawaji Formation												
KK65*	B	12	146.1	−61.4	134.3	−65.1	197.8	3.1	36°58.5′	139°39.8′	RWT	6.8
KK67*	C	11	166.6	−50.9	164.2	−24.1	113.0	4.3	36°56.5′	139°37.9′	RWT	
KK69*	C	12	163.1	−54.0	163.2	−26.0	76.2	5.0	36°56.2′	139°38.5′	RWT	
KN70*	B	12	148.4	−58.8	76.4	−84.3	306.0	2.5	36°58.5′	139°39.7′	RWT	
KN72*	C	12	163.1	−53.5	151.8	−31.4	69.1	5.3	36°55.8′	139°39.6′	RWT	
KN74*	C	13	143.6	−55.8	153.1	−51.8	341.7	2.2	36°56.2′	139°39.1′	RWT	
Kinugawa Formation												
KN61	B	12	194.5	−63.1	194.5	−63.1	479.3	2.0	36°52.6′	139°26.3′	RWT	6.7
KN63	B	11	208.5	−46.7	205.3	−55.4	375.0	2.4	36°52.9′	139°25.9′	RWT	
KN65	B	12	192.0	−53.1	178.9	−57.4	366.3	2.3	36°53.3′	139°25.7′	RWT	
KN67	B	10	204.6	−55.4	203.1	−49.6	158.4	3.9	36°50.2′	139°27.1′	RWT	
KN81	B	5	179.4	−65.7	186.6	−56.5	61.1	9.9	36°52.0′	139°31.8′	RWT	7.8
Katashinagawa Formation												
Northeast												
KK24	A	10	177.0	−73.9	210.3	−60.2	144.3	4.0	36°47.6′	139°19.9′	RWT	8.9
KK12	B	4	145.5	−78.3	189.3	−44.8	203.2	6.5	36°48.9′	139°20.0′	RWT	9.1
KK20	A	9	186.6	−76.7	209.5	−61.7	99.5	5.2	36°48.9′	139°19.7′	RWT	7.7
KK22	A	12	211.2	−68.6	221.1	−49.4	217.8	3.0	36°48.9′	139°19.7′	RWTt	
KN21	B	12	313.0	81.2	20.6	70.8	94.3	4.5	36°52.4′	139°18.6′	RWT	8.9
Southwest												
KK02*	B	8	15.7	61.7	7.8	59.5	99.7	5.6	36°42.1′	139°13.9′	DRWT	33.8
KK03	A	10	1.2	50.6	1.2	50.6	354.1	2.6	36°41.5′	139°14.2′	DRWT	
KK08	A	12	5.1	48.3	352.3	43.9	136.5	3.7	36°43.2′	139°15.3′	RWT	13.3
KK10	A	10	3.4	51.2	359.4	47.5	277.6	2.9	36°43.1′	139°15.5′	RWT	
KK14	A	11	354.9	47.7	0.0	31.8	347.3	2.5	36°42.9′	139°15.7′	RWT	13.7
KK16	A	9	20.3	52.6	17.7	54.9	251.4	3.3	36°43.0′	139°16.7′	RWT	12.0
Okkai Formation												
KK04	B	10	165.6	−35.9	167.4	−60.8	208.4	3.4	36°42.5′	139°12.7′	AWT	9.8
KK18	B	11	181.0	−30.9	183.6	−27.0	80.3	5.1	36°42.2′	139°12.7′	AWT	10.9
Ashio Formation												
KN15	B	12	29.0	33.8	30.1	48.8	101.2	4.3	36°39.7′	139°25.6′	DWT	10.2
KN17	C	10	18.7	18.9	18.7	18.9	433.7	2.3	36°39.2′	139°24.9′	DWT	
KN19	A	12	15.2	25.3	15.9	12.4	150.5	3.5	36°38.6′	139°25.1′	DWT	

Formation and at 500°C for the Ashio Formation, the high-temperature component reveals straightforward demagnetization behaviour with an unblocking temperature of about 600°C. Thermomagnetic analysis revealed the presence of magnetic minerals with a Curie temperature of 350°C in KN725 of the Kawaji Formation, suggesting that the middle-temperature component is carried by maghemite or pyrrhotite. The high-temperature component resides probably in titanium-poor magnetite. The highest-temperature component direction is recognized as the characteristic direction.

4.2. Palaeomagnetic directions

Palaeomagnetic directions of the low-temperature component magnetization removed by 250°C in the categories B and C are dominated in situ by a component in the present direction of the Earth's magnetic field (Fig. 3). This component is therefore ascribable to viscous partial thermoremanent magnetization acquired in recent times (Pullaiah et al., 1975).

The high-temperature component directions are clustered tightly in each site (Table 3): precision parameter ranges from 55.5 to 851.9. Three formations (the Osawa, Yatakezawa and Ashio Formations) revealed normal polarity, and three formations (the Kawaji, Kinugawa, and Okkai Formations) reversed ones. Both polarities are observed in the Katashinagawa Formation.

A formation mean direction for the high-temperature component magnetizations is calculated for each formation (Table 4). According to difference in lithologic features or the fission-track ages, the Yatakezawa and Katashinagawa Formations are divided into two groups: north and south areas. Palaeomagnetic data of site KK02 with an age of 33.8 Ma are not incorporated into the data set of the southern part of the Katashinagawa Formation. The seven formations except for the Kawaji Formation show good grouping before and after tilt correction ($16<k$). Although the precision parameter decreases after tilt correction for the Osawa and Ashio Formations, the decrease is insignificant for a negative fold test even at 95% level. Increase in the dispersion is possibly due to uncertainty in strike measurements (of the order of $\pm 5°$) for estimating the palaeohorizontal plane from the eutaxitic structure in the welded tuffs. The high-temperature component after tilt correction, except for the Kawaji Formation, is recognized as a characteristic palaeomagnetic direction for each site.

The mean palaeomagnetic direction for the high-temperature component magnetizations of the Kawaji Formation reveals a significant decrease from 128 to 10 in the precision parameter after tilt correction (Fig. 4). A similar decrease in the precision parameter from 119 to 11 is observed for the mean palaeomagnetic direction of the low-temperature component (Table 4). The ratio of the precision parameter before and after tilt correction gives a negative fold test at the 99% level (McElhinny, 1964) for both the low- and high-temperature components. These indicate that the high-temperature component is a secondary magnetization acquired after tilting, the same as the low temperature one. Occurrence of magnetic minerals with a Curie temperature of 350°C in this formation is presumably evidence of origination of secondary magnetization from a rock magnetic view point. A fairly large counter-clockwise deflection in declination with reversed polarity of the palaeomagnetic direction in situ for the Kawaji Formation is also observed in the tilt-corrected direction of the Shiobara Formation, implying that the Kawaji Formation acquired its secondary magnetization when the Shiobara Formation was produced at 5.0 Ma. The palaeomagnetic directions of the Kawaji Formation are, therefore, excluded from further tectonic investigation.

A mean palaeomagnetic direction on the basis of the data set from 29 sites except for the Kawaji Formation reveals an increase in the precision parameter

Note to Table 3: The magnetization directions of the high-temperature component are listed. Category A: single-component magnetizations with high-temperature component; B: distinctive two-component magnetizations; C: distinctive three-component magnetizations. N is the number of field oriented samples. D and I are declination and inclination respectively; k is the Fisher precision parameter; α_{95} is the radius of the cone of 95% confidence within site. Ages are calculated by fission-track analysis in this study. Rock type: AWT = andesite welded tuff; DWT = dacite welded tuff; DRWT = dacite–rhyolite welded tuff; RWT = rhyolite welded tuff.

[a] Age assignments for the Osawa Formation after Kitamura (1986).

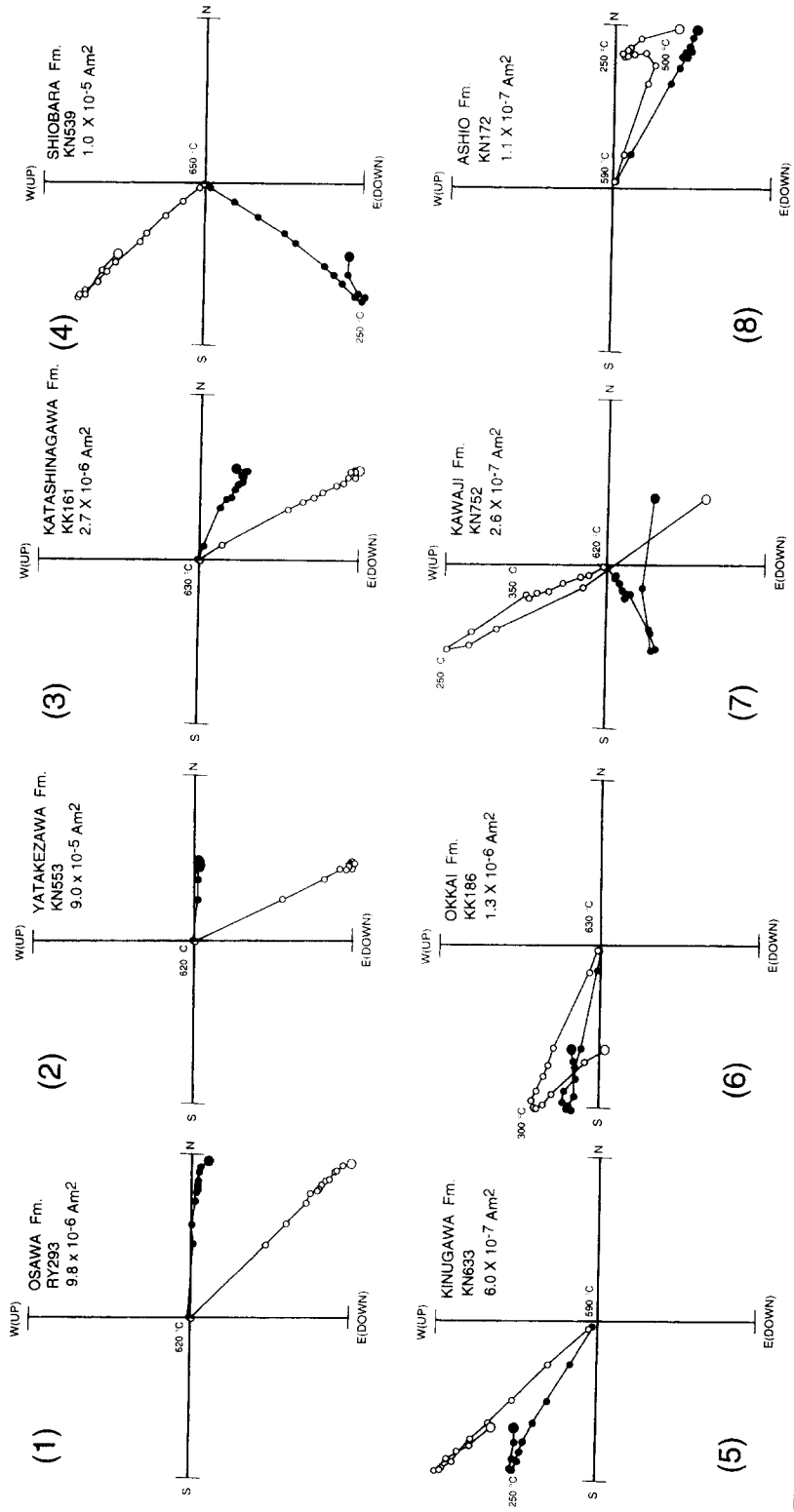


Fig. 3. Orthogonal projections of magnetization vector end points during thermal demagnetization experiments for specimens of three categories (see text). Category A (1–3: RY29, KN55, KK16), category B (4–6: KN53, KN63, KK18), and category C (7, 8: KN75, KN17). Solid symbols for horizontal projection, open symbols for vertical projection. Tectonic tilt correction has not been applied.

Table 4
Mean palaeomagnetic directions for the Miocene welded tuffs from the northern part of central Japan

Formation	N	In situ				Tilt corrected				Locality	
		D (°)	I (°)	k	α_{95} (°)	D (°)	I (°)	k	α_{95} (°)		
(1) *Formation mean*											
Osawa	3	352.3	43.3	89.5	13.1	358.6	41.4	25.7	24.8	37°55′N	140°12″E
Yatakezawa											
North	2	5.1	66.4			348.5	69.3			37°08′N	139°32′E
South	2	4.4	60.3			27.9	47.5			37°08′N	139°32′E
Shiobara	2	305.8	32.5			318.9	44.6			36°59′N	139°47′E
Kawaji											
High	6	335.9	56.4	128.4	5.9	334.4	47.8	9.5	22.9	36°56′N	139°39′E
Low	5	357.4	53.9	118.8	7.1	353.4	55.1	11.1	24.0	36°56′N	139°39′E
Kinugawa	5	18.7	57.0	82.3	8.5	14.1	56.8	106.9	22.9	36°52′N	139°26′E
Katashinagawa											
Northeast	5	358.2	77.4	84.2	8.4	26.0	58.0	42.1	11.9	36°45′N	139°20′E
Southwest	5	4.7	50.4	166.6	6.0	1.4	46.0	57.4	10.1	36°42′N	139°13′E
Okkai	2	353.5	33.6			358.0	44.2			36°42′N	139°13′E
Ashio	3	21.0	26.0	64.9	15.4	20.5	26.6	16.0	31.9	36°36′N	139°24′E

(2) *Mean for the northern part of central Japan* (except for the seven sites with * in Table 3)

	N	In situ				Tilt corrected				Locality 36.8°N 139.3°E V G P		
		D (°)	I (°)	k	α_{95} (°)	D (°)	I (°)	k	α_{95} (°)	Lat. (°N)	Long. (°E)	A_{95} (°)
15.5–5.0 Ma	29	0.5	53.6	14.7	7.2	7.3	50.3	17.2	6.6	82.6°N	259.1°E	7.3
15.5–12.0 Ma	8	359.6	47.5	78.9	6.3	0.2	43.9	43.5	8.5	79.3°N	315.7°E	8.9
10.9–5.0 Ma	21	1.0	56.1	11.4	9.8	10.7	52.7	14.8	8.6	81.2°N	234.9°E	9.5

(1) Formation mean palaeomagnetic directions. The formation mean directions are calculated for the high-temperature component in situ and after tilt correction except for the Kawaji Formation. Both mean directions for the high- and low-temperature component are listed for the Kawaji Formation. *N* is the number of field-oriented samples. *D* and *I* are declination and inclination respectively; *k* is the Fisher precision parameter; α_{95} is the radius of the cone of 95% confidence within site.
(2) A mean direction for the northern part of central Japan. The mean direction for the northern part of central Japan is obtained as a value at a representative point in of northern part of northeast Japan (36.8°N, 139.3°E) for 29 sites. Data denoted by asterisk in Table 3 are not used for calculation of the mean direction. Mean directions are also calculated on the basis of data sets for two periods: 15.5–12.0 Ma and 10.9 Ma–5.0 Ma. VGP is virtual geomagnetic pole. A_{95} is radius of the circle of 95% confidence about the mean pole.

from 14.7 to 17.2 (Fig. 5). The mean direction after tilt correction ($D = 7.3°$, $I = 50.3°$, $\alpha_{95} = 6.6°$) is recognized to be a characteristic palaeomagnetic direction in Middle to Late Miocene time for the northern part of central Japan.

5. Discussion

Combining the previously obtained chronological data with our new fission-track ages, Miocene formations for our palaeomagnetic study cover an age range between 15.5 Ma and 5.0 Ma. The study area covers 140 × 50 km in latitudinal and longitudinal directions.

The palaeomagnetic direction of this period for the northern part of central Japan reveals a northerly direction (Fig. 6 and Table 4): Declinations for the periods of 15.5–12.0 Ma and 10.9–5.0 Ma are 0.2° and 10.7°, respectively. The mean direction of 29 sites after tilt correction for a period between

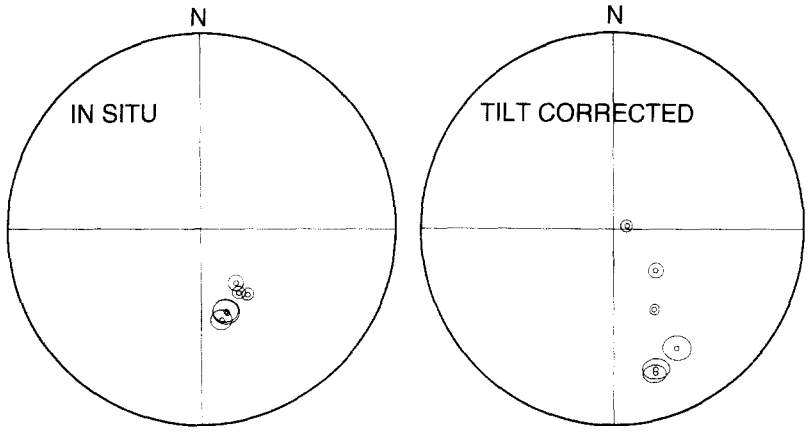


Fig. 4. Equal-area projections of the site mean direction with a circle of 95% confidence for the Kawaji Formation, before and after tilt correction. Fisherian precision parameter k reduced from $k = 128.4$ to $k = 9.5$ after tilt correction.

15.5 Ma and 5.0 Ma is $D = 7.3°$, $I = 50.3°$ ($\alpha_{95} = 6.6°$) at a reference point (lat. = 36.8°N, long. = 139.3°E). Effect of the geomagnetic field secular variation is excluded because of the long time average which is supported by chronology and through the presence of the normal and reversed polarities. The northerly direction for Miocene times was reported in the neighbouring areas, although these data were obtained without tilt correction. Four sites of dyke rocks (10.1–13.7 Ma) from the Minamiaizu–Koriyama area and the dyke rocks (12–13.6 Ma) in the Niigata area also show northward directions of $D = -3.8°$, $I = 56.7°$, $\alpha_{95} = 6.7°$, and $D = 3.8°$, $I = 52.4°$, $\alpha_{95} = 4.5°$, respectively (Hirooka et al., 1986). Little deflection in declination characterizes the Miocene palaeomagnetic direction between 15.5 Ma and 5.0 Ma for the northern part of central Japan.

This Miocene palaeomagnetic direction is compared with the palaeomagnetic direction of the northeastern Japan block for Miocene times. After 15 Ma palaeomagnetic directions are characterized by a northerly direction in the northeastern Japan block from Yamagata to Oshima Peninsula of west-

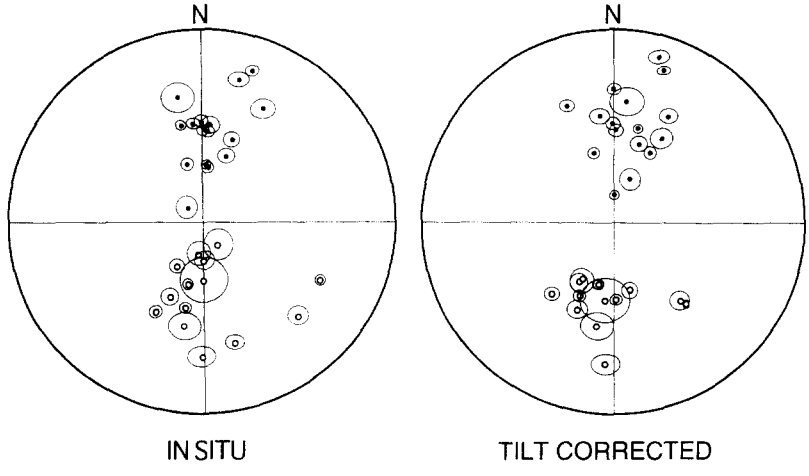


Fig. 5. Equal-area projections of the site mean direction with a circle of 95% confidence based on the data from the Osawa, Yatakezawa, Shiobara, Kinugawa, Katashinagawa, Okkai, and Ashio Formations, before and after tilt correction. Fisherian precision parameter k and the radius of cone of 95% confidence α_{95} changes from ($k = 14.7$, $\alpha_{95} = 7.2°$) to ($k = 17.2$, $\alpha_{95} = 6.6°$) for 29 sites.

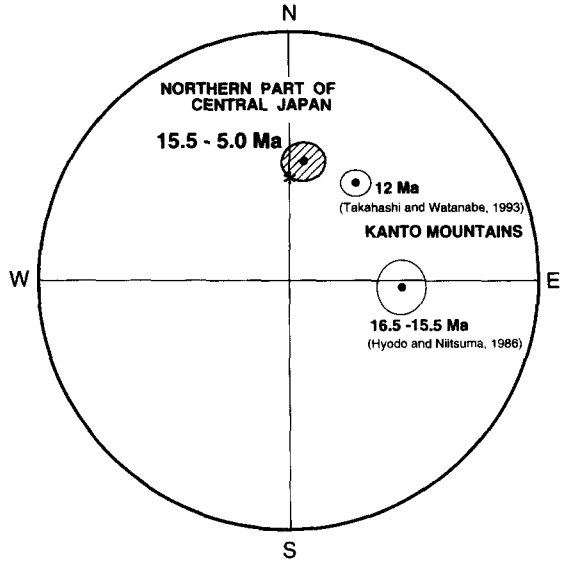


Fig. 6. Mean palaeomagnetic field directions for the northern part of central Japan (this study, shaded) and the Kanto Mountain region (16.5–15 Ma and 12 Ma) (Hyodo and Niitsuma, 1986; Takahashi and Watanabe, 1993) with circles of 95% confidence. Asterisk: present geocentric axial dipole field direction at a representative point in the northern part of northeast Japan (36.8°N, 139.3°E).

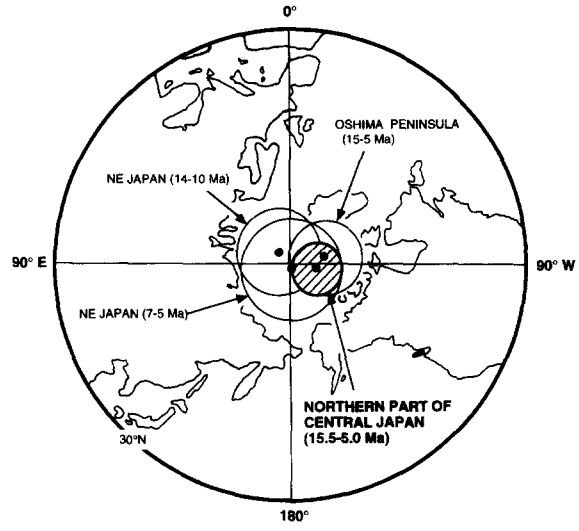


Fig. 7. Miocene palaeomagnetic poles (with associated circles of 95% confidence) of the northern part of central Japan (this study), northeast Japan (14–10 Ma and 7–5 Ma) (Otofuji et al., 1985b, 1994) and the Oshima Peninsula of Hokkaido Island (15–5 Ma) (Tanaka et al., 1991). Polar stereographic projection.

ern Hokkaido, although a westerly declination prevails before 15 Ma (Otofuji et al., 1985b; Otofuji et al., 1994). Palaeomagnetic poles for after 15 Ma are plotted in Fig. 7. Palaeomagnetic poles of northeast Japan are 85.4°N, 43.3°E and 88.3°N, 186.7°E for the periods of 14–10 Ma and 7–5 Ma, respectively (Otofuji et al., 1994). Tanaka et al. (1991) reported for 15–5 Ma a palaeomagnetic pole of 80.1°N, 280.5°E, $A_{95} = 10.4°$ for the Oshima Peninsula of western Hokkaido. The palaeomagnetic pole of the northern part of central Japan between 15.5 Ma and 5.0 Ma falls onto these poles of northeast Japan. This strongly suggests that the northern part of central Japan and northeast Japan behaved as a unified block since 15 Ma. These palaeomagnetic pole positions are close to the present rotation pole, indicating that these blocks have been subjected to little rotation and translation after 15 Ma.

In contrast, a large difference in palaeomagnetic directions is manifested between the northern part of central Japan and the Kanto Mountain region. Largely an easterly declination has been observed from the Miocene rocks of the Chichibu area in the Kanto Mountain region to the southwest of our study area (see Figs. 1 and 6): $D = 94°$ from Neogene (16.5–15.5 Ma) sedimentary rocks (Hyodo and Niitsuma, 1986) and $D = 35°$ from igneous rocks of the Yaekubo Formation with ages of ca. 12 Ma (Takahashi and Watanabe, 1993). The difference in declination between the two blocks implies a large relative block movement later than 15 Ma. The Kanto Mountain region rotated with respect to the northern part of central Japan through more than 60° during the period between 16.5–15.5 Ma and 12 Ma, and the successive rotation through about 30° continued since then.

Prior to 16 Ma, differential rotation had already occurred between the northern part of central Japan and the Kanto Mountain region. The Oligocene–Early Miocene rocks from Sado Island and the Fukushima area reveal counter-clockwise deflection in declination as $D = 333°$, and $D = 346°$, respectively (Hirooka et al., 1986). This palaeomagnetic evidence implies that the northern part of central Japan was subjected to counter-clockwise rotation more than 14° later than 20 Ma when the Kanto Mountain region rotated clockwise as the easternmost edge of southwest Japan during opening of the

Japan Sea (Hyodo and Niitsuma, 1986). The present study shows that the earlier differential rotation of the two blocks has continued until at least 12 Ma.

The northern part of central Japan, however, has been united with the Kanto Mountain region at least by 6 Ma, because a northerly direction is reported ($D = 3.5°$, $I = 64°$) for the 6 Ma palaeomagnetic direction of the Chichibu quartz diorite of the Kanto Mountain (Takahashi and Nomura, 1989).

Palaeomagnetic data have established that a major tectonic boundary was located between the northern part of central Japan and the Kanto Mountain region in Middle Miocene times. The boundary was between the sampling area of the Katashinagawa Formation with an age of 13 Ma (36°45′N, 139°15′E) and that of the Yaekubo Formation with 12 Ma (36°15′N, 138°34′E) (Fig. 1). A pronounced low gravity anomaly zone with a width of about 30 km extends from southeast to northwest, ranging from the Kanto Plain, through the area between the northern part of central Japan and the Kanto Mountain, to the Japan Sea (Komazawa et al., 1992; Gravity Research Group in Southwest Japan, 1994; Shichi and Yamamoto, 1994; Komazawa et al., 1994) (Fig. 8). Proximity of this anomaly zone and the tectonic boundary predicted by the palaeomagnetism suggest that the block boundary has remained as a low Bouguer anomaly zone. Minimum Bouguer anomaly in the low zone is less than 0 mGal, whereas the maximum at the northern part of central Japan and the Kanto Mountain is up to 90 mGal. The fairly large low anomaly with the short wavelength is explained by crustal thickening of low-density material which is formed as a result of compression of lithosphere between these two blocks.

The presence of a structural discontinuity is also inferred from geological features between the northern part of central Japan and the Kanto Mountain region. Eastward offsets by 100 km of the volcanic front of Middle Miocene times has been provoked along the low gravity anomaly zone (Seno and Maruyama, 1984; Takeuchi, 1986). The block boundary is, therefore, recognized as a fundamentally transcurrent fault, along which the northern part of central Japan has been transported southeastward dextrally. Geoscientific studies of palaeomagnetism, gravity and geology suggest that the northern part of central Japan and the Kanto Mountain were bordered by a dextral trans-compressional fault system at least until 12 Ma.

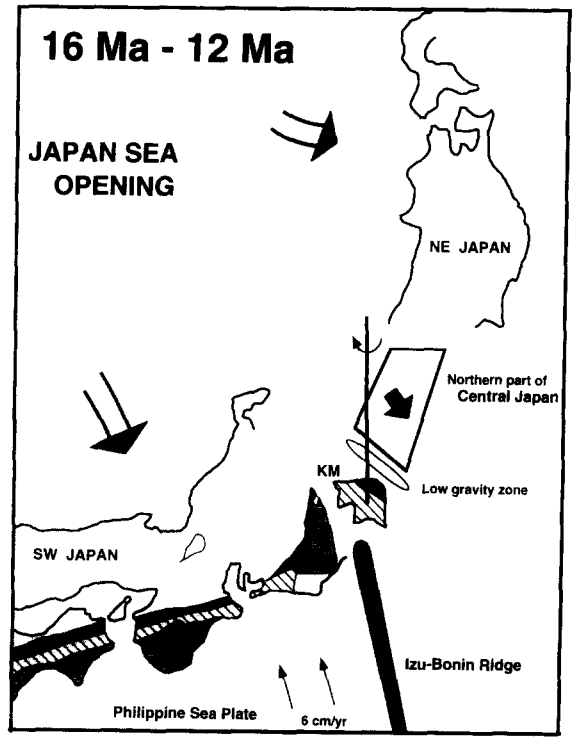


Fig. 8. Tectonic setting at the latest stage of the Japan Sea opening between 16 Ma and 12 Ma. After the easternmost part of southwest Japan collided with the Izu-Bonin ridge in the latest stage of clockwise rotation of southwest Japan, a significant change appeared in the trend of the lineation of the zonal structure of the Outer zone between central Japan and the western part of southwest Japan and the Kanto Mountain region (KM). The Kanto Mountain region was rotated clockwise through 60° between 16.5 Ma and 12 Ma. The rotation is attributed to the torque which arose by the north-northwestward motion of the Izu-Bonin ridge and the gradually southeastward motion of the northern part of central Japan. Compressive regime with dextral movement predominated between the northern part of central Japan and the Kanto Mountain region, so that the narrow and low-gravity zone was produced.

The southeastward movement of the northern part of central Japan plays an important part in rotation of the Kanto Mountain region (Fig. 8). The eastern part of southwest Japan had reached the Izu-Bonin ridge on the Philippine Sea plate at a later stage of the clockwise rotation, later than 16.5 Ma (Seno and Maruyama, 1984; Niitsuma, 1989; Otsuki, 1990). The block ranging from the Kanto Mountain region to Boso Peninsula, which is the easternmost part of

southwest Japan, collided with the Izu-Bonin ridge and began to rotate through 60° between 16.5 Ma and 12 Ma. The rotation is ascribed to the force exerted from dextral motion of the northern part of central Japan as well as to north-northwestward motion of the Izu-Bonin ridge.

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